NLP

22-06-26

Attention Mechanism :

ANN → RNN → LSTM → Attention

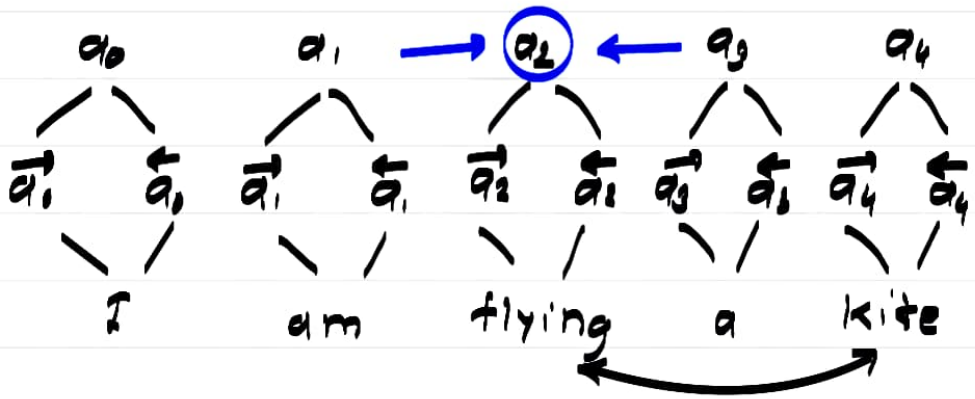
2017 : Attention is All you need!
Google

Attention mechanism introduced a completely new way of capturing context in NLP tasks.

A new family of models called Transformers was created using attention mechanism.

Transformer Architecture:

- Transformer models use attention to capture context of the sequence.



$$a_2 = \vec{w}\,\vec{a_2} + \overleftarrow{w}\,\overleftarrow{a_2}$$

$$\vec{a_2} = \vec{w}_x x_2 + \vec{w}_a \vec{a_1} + \vec{b}$$

$$\overleftarrow{a_2} = \overleftarrow{w}_x x_2 + \overleftarrow{w}_a \overleftarrow{a_3} + \overleftarrow{b}$$

The sequence context is being passed to a_2 using only two vectors → $\vec{a_1}$ and $\overleftarrow{a_3}$

$\vec{a_1}$ ↓ left side context

$\overleftarrow{a_3}$ ↓ right side context

In attention, the context is passed using all the vectors representing the sequence.

In attention, each word uses the vector of every other word in given sequence to capture its complete meaning.

I am flying a kite

↓ tokenization - word piece tokenization

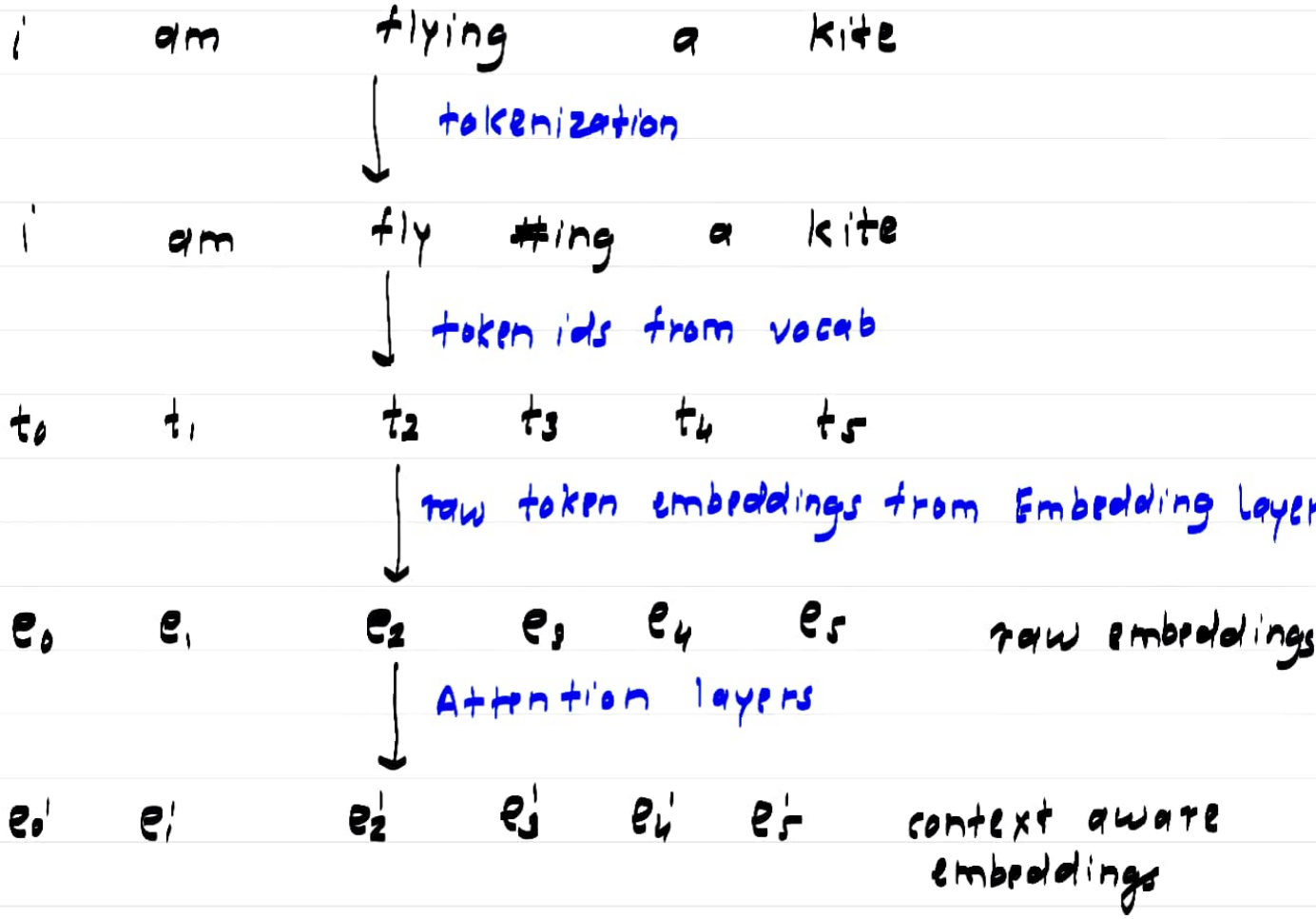
i am fly #ing a kite

↓ token ids from vocabulary

t_0 t_1 t_2 t_3 t_4 t_5

We randomly initialize the word embeddings for every token in vocabulary

This is done using an Embedding layer.



Raw embedding :- Individual embedding of every token
- doesn't have information about any other token in the sequence

context aware embedding:
- context aware vector of every token
- has information about all other tokens in the sequence

Embedding Layer:

- it stores raw embedding for every token in the vocabulary

vocab size = 20k tokens

embedding layer matrix

$$E = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ \vdots \\ v_{20k} \end{bmatrix}_{20k \times d}$$

dim of every vector = d

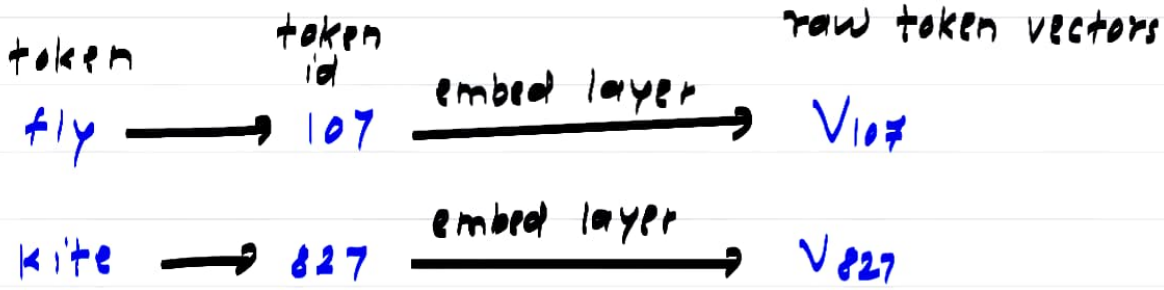
$$v_1 = [v_{11} \; v_{21} \; v_{31} \; v_{41} \; \ldots \; v_{d1}]$$

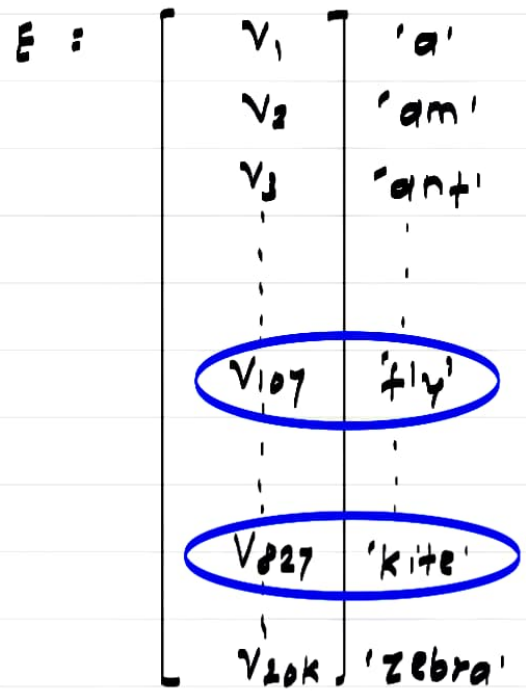
$$v_n = [v_{1n} \; v_{2n} \; v_{3n} \; v_{4n} \; \ldots \; v_{dn}]$$

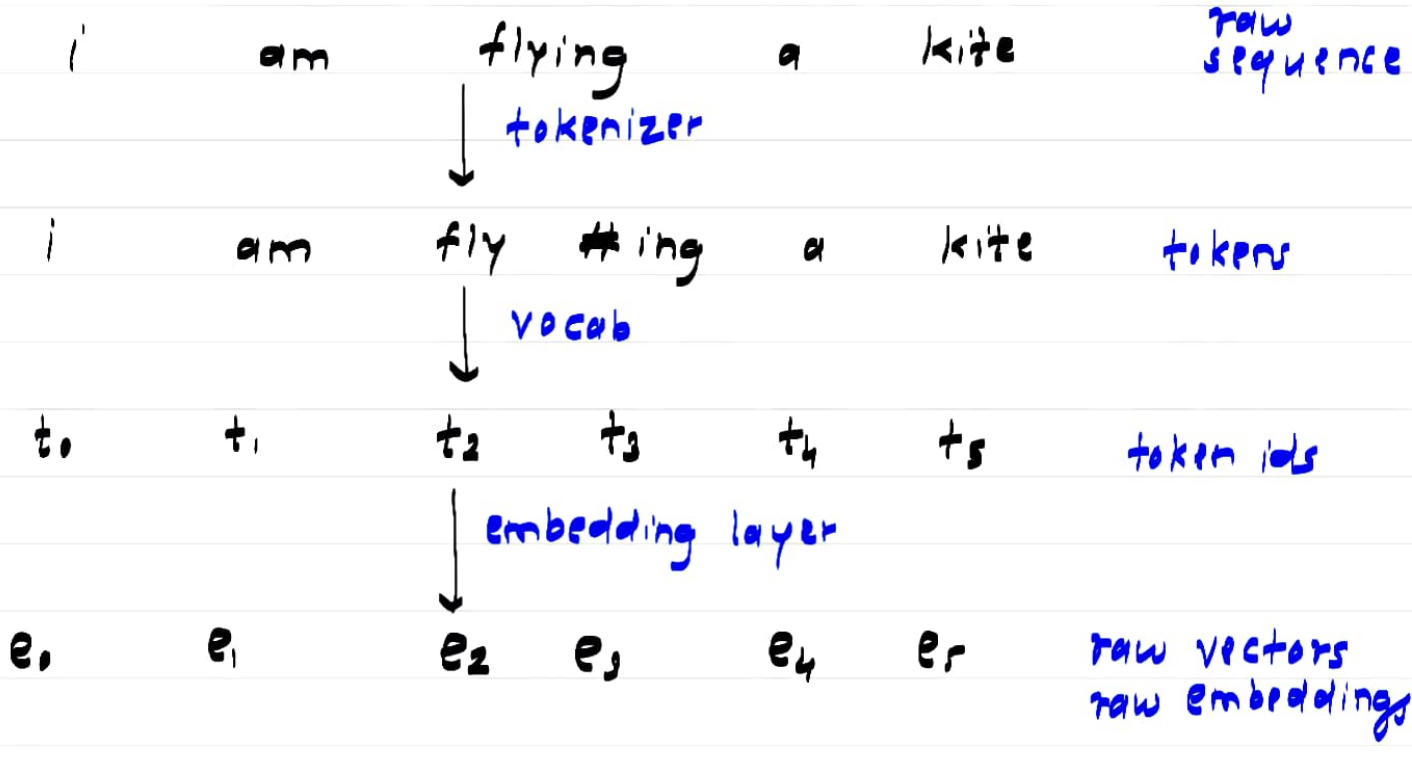
- At the start of training matrix E is initialized randomly
- During training the matrix E is updated
- matrix E acts as weight matrix for embedding layer

matrix E is model parameter in transformer which gets updated during training

vector dimension d is a hyperparameter


token
token id
raw token vectors
fly
107
embed layer
V_{107}
kite
827
embed layer
V_{827}


E :
V_1 'a'
V_2 'am'
V_3 'ant'
V_{107} 'fly'
V_{827} 'kite'
V_{10k} 'zebra'


I am flying a kite
raw sequence
tokenizer
i am fly #ing a kite
tokens
vocab
t_0 t_1 t_2 t_3 t_4 t_5
token ids
embedding layer
e_0 e_1 e_2 e_3 e_4 e_5
raw vectors
raw embeddings

Attention mechanism is applied on raw embeddings

Before this we capture the positional information of every token as positional encoding

raw vector dim : d
posional encoding vector dim : d

eg.

raw vector dim : 6

sequence →	t_0	t_1	t_2	t_3
raw vectors →	e_0	e_1	e_2	e_3

$$e_1 : [e_{01} \quad e_{11} \quad e_{21} \quad e_{31} \quad e_{41} \quad e_{51}]$$

$$p_1 : [p_{01} \quad p_{11} \quad p_{21} \quad p_{31} \quad p_{41} \quad p_{51}]$$

$$P(2i, pos) = \sin\left(\frac{pos}{10000^{2i/d}}\right)$$

$$P(2i+1, pos) = \cos\left(\frac{pos}{10000^{2i+1/d}}\right)$$

$$e_1 : [0.1 \quad 0.45 \quad -0.76 \quad -0.5 \quad -1 \quad 1.2]$$

$$p_1 : [p_{01} \quad p_{11} \quad p_{21} \quad p_{31} \quad p_{41} \quad p_{51}]$$

$$p_{01} = \sin\left(\frac{1}{10000^{0/6}}\right) \quad p_{11} = \cos\left(\frac{1}{10000^{1/6}}\right) \quad p_{21} = \sin\left(\frac{1}{10000^{2/6}}\right)$$

$$P_{31} = \cos\left(\frac{1}{10000^{3/6}}\right)$$

i am fly #ing a kite

↓ raw embeddings

e_0 e_1 e_2 e_3 e_4 e_5

+

p_0 p_1 p_2 p_3 p_4 p_5

↓

$\bar{e}_0$ $\bar{e}_1$ $\bar{e}_2$ $\bar{e}_3$ $\bar{e}_4$ $\bar{e}_5$

position aware embeddings

$e_1 = [e_{01} \quad e_{11} \quad e_{21} \quad e_{31} \quad e_{41} \quad e_{51}]$

$p_1 = [p_{01} \quad p_{11} \quad p_{21} \quad p_{31} \quad p_{41} \quad p_{51}]$

$\bar{e}_1 = e_1 + p_1$

$= [e_{01}+p_{01} \quad e_{11}+p_{11} \quad e_{21}+p_{21} \quad e_{31}+p_{31} \quad e_{41}+p_{41} \quad e_{51}+p_{51}]$

In attention mechanism, positional encodings are needed to capture the order of words.

This is essential as tokens are not processed sequencially in attention. Tokens are processed at the same time parallely.

Changing the position of token in sequence changes it position aware embedding.

$$e_i' = f(\bar{e}_0, \bar{e}_1, \bar{e}_2, \bar{e}_3, \bar{e}_4, \bar{e}_5)$$

all vectors in a sequence will be used for generating context aware embeddings.